Digital Logic Design

PRE-WORK
SEMICONDUCTOR IC DESIGN FOUNDATION
AND READINESS PROGRAMME

2-Hour Training Module

Module 3 at a glance

ORIENTATION

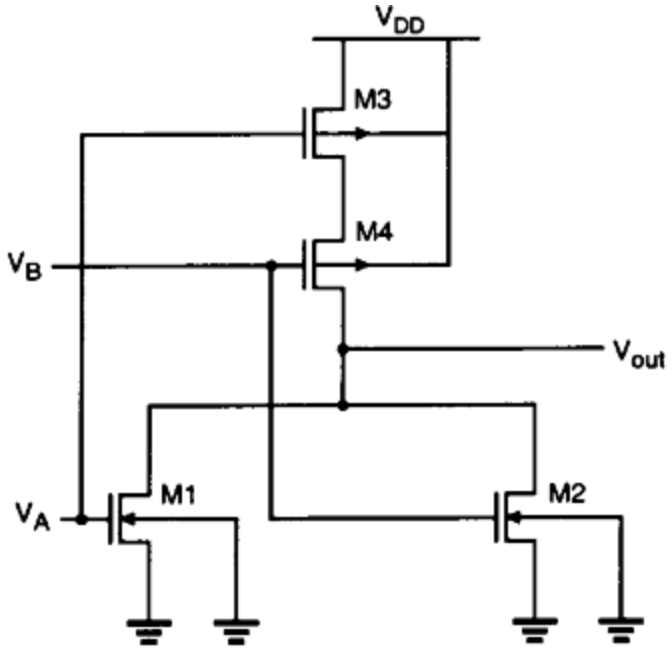
Develop participants' ability to understand how digital logic elements are combined into functional digital systems. Participants will learn to analyse circuit behaviour, signal flow and timing before progressing into HDL-based IC design.

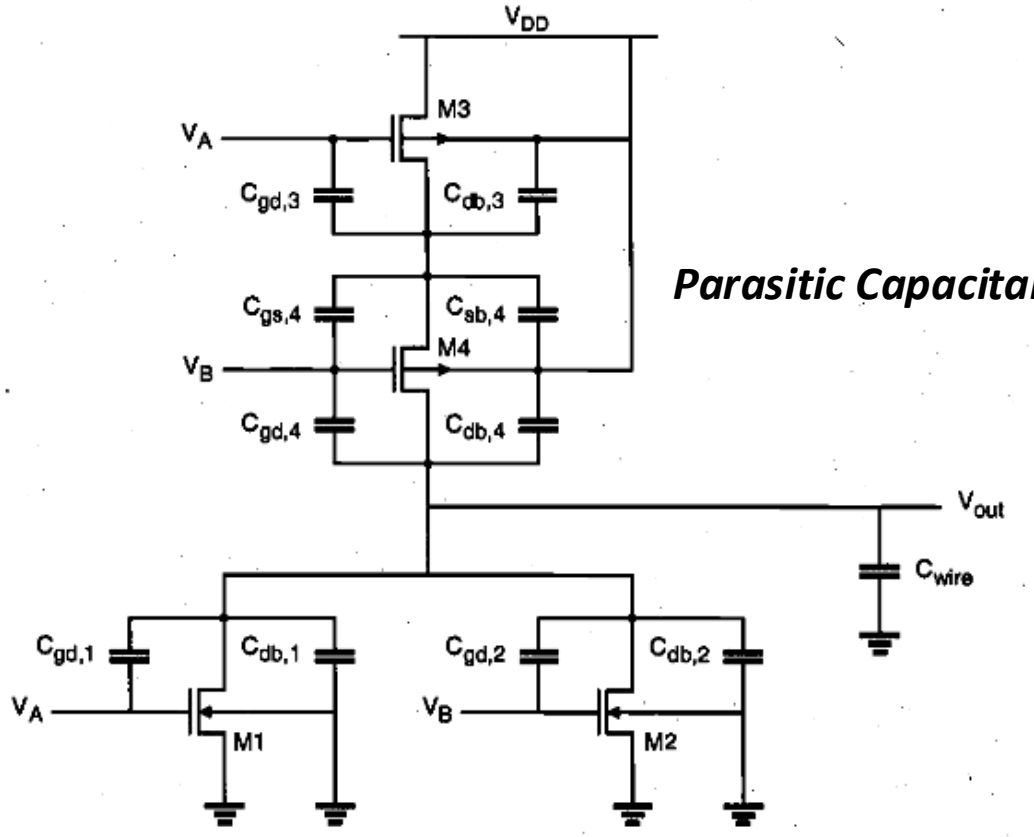
Learning outcomes

ORIENTATION

Reading simple digital circuit diagrams; designing basic combinational circuits;designing simple sequential circuits; understanding inputs, outputs and signal flow; interpreting timing diagrams; understanding propagation delay; recognising synchronous and asynchronous behaviour; understanding basic hardware description concepts.

NOR gate propagation delay





Parasitic Capacitance

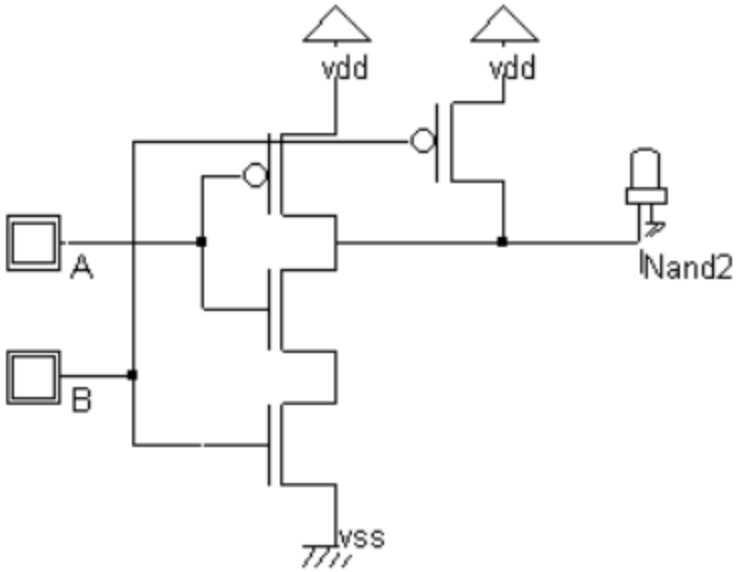
Propagation Delay

$$\tau_{PHL} = \frac{C_{load}}{k_n\left(V_{DD} - V_{T,n}\right)}\left[\frac{2V_{T,n}}{V_{DD} - V_{T,n}} + ln\left(\frac{4\left(V_{DD} - V_{T,n}\right)}{V_{DD}} - 1\right)\right]$$

Equivalency

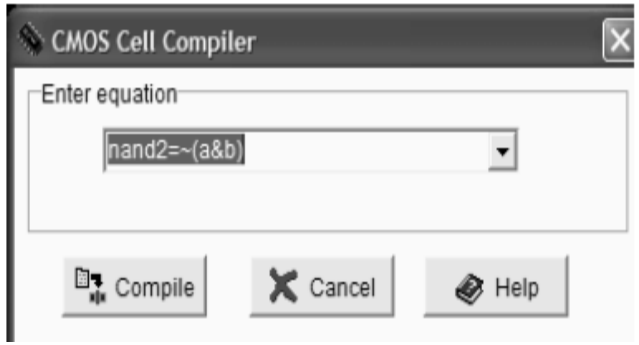
$$\tau_{PLH} = \frac{C_{load}}{k_P\left(V_{DD} - |V_{T,p}|\right)}\left[\frac{2|V_{T,p}|}{V_{DD} - |V_{T,p}|} + ln\left(\frac{4\left(V_{DD} - |V_{T,p}|\right)}{V_{DD}} - 1\right)\right]$$

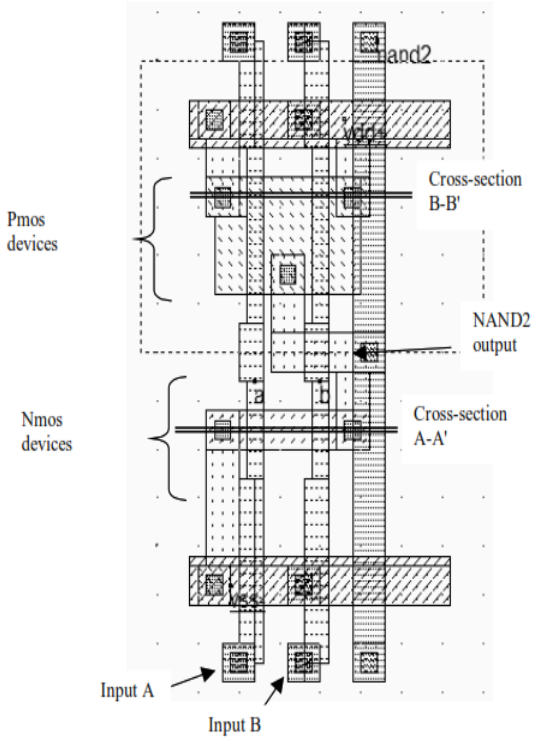
NAND gate EXERCISE 15 mins (pg 43-44 Manual)



the CMOS NAND gate design (NandCmos.SCH)

You may load the NAND gate design using the command File → Read→NAND.MSK. Or, to compile directly the NAND gate into layout with Microwind2.





Full Adder EXERCISE 15 mins (pg-59-62 Manual)

Full Adder

A	B	C	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

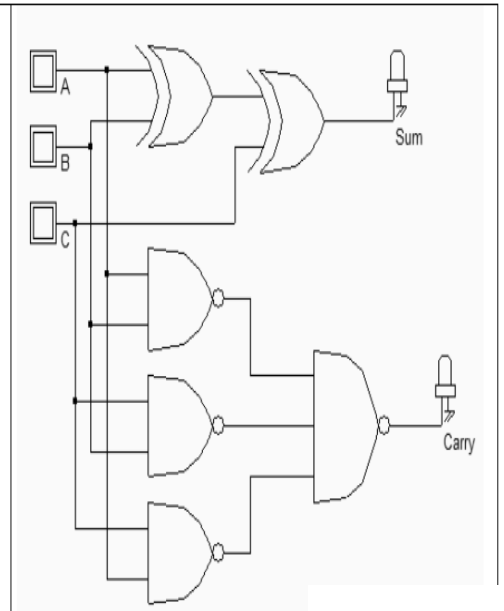


Fig. 6-6. The truth table and schematic diagram of a full-adder(FADD.SCH)

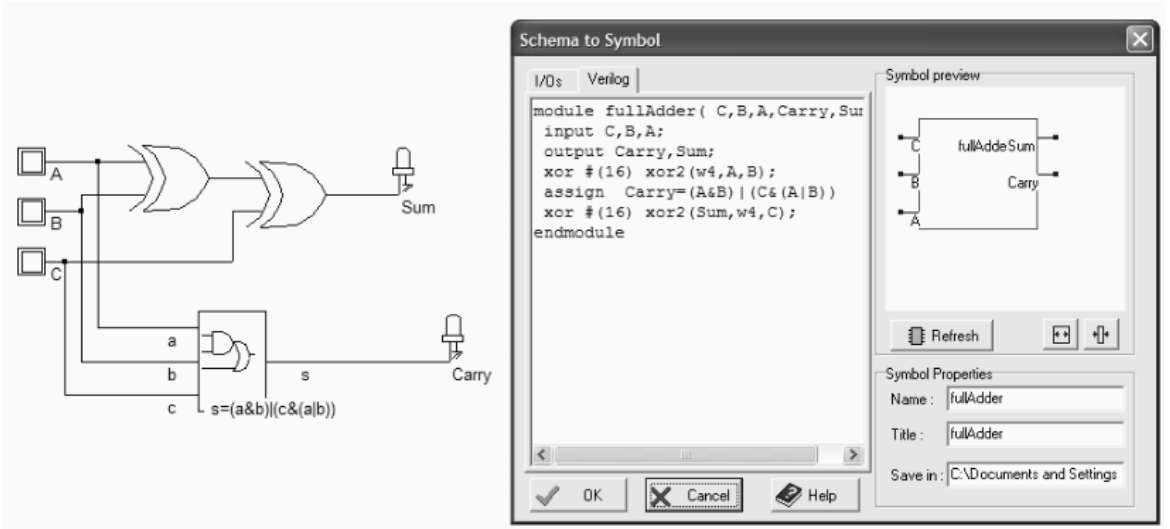
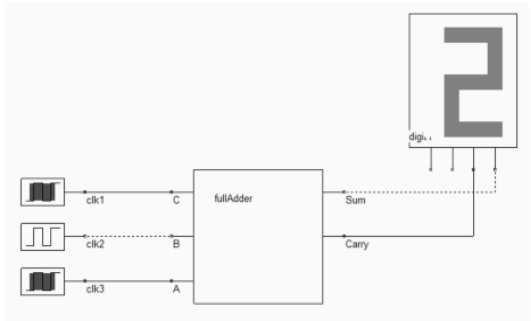


Fig. 6-7 Verilog description of the full adder (FullAdder.SYM)



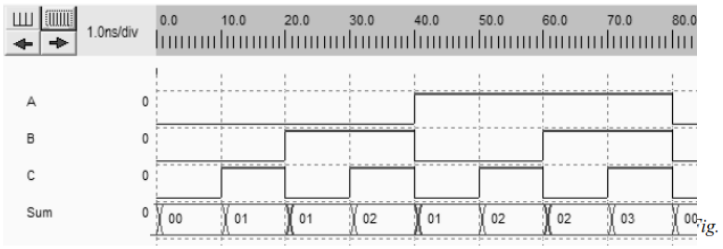


Fig. 6-8. Testing the new Adder symbol using clocks (FaddTest.SCH)

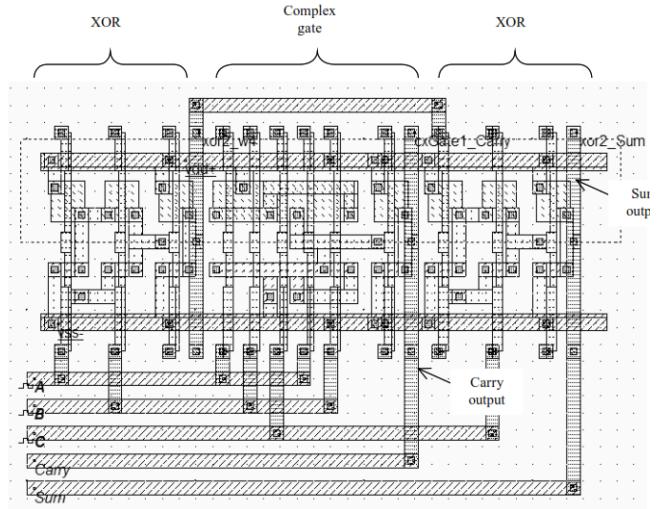


Fig. 6-9. The full adder compiled from the Verilog description (FullAdder.MSK)

Counter EXERCISE 15 mins (pg 78- Manual)

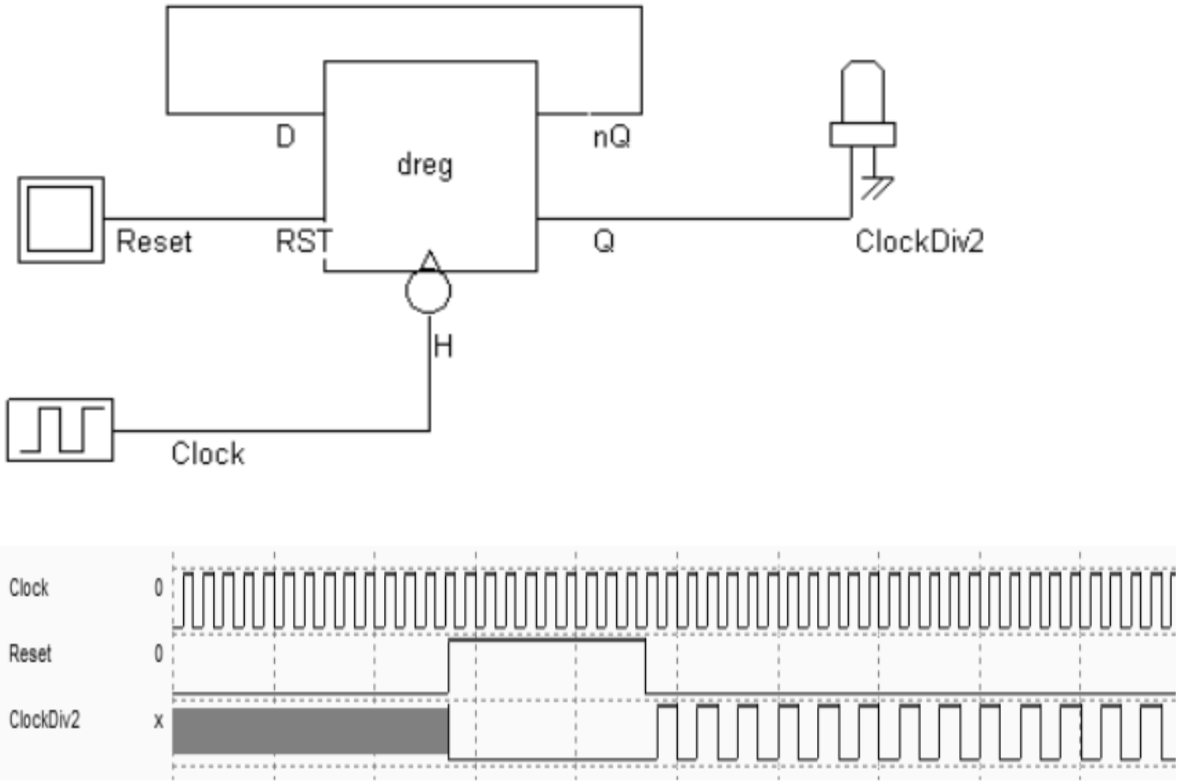


Fig. 7-14. Schematic diagram and simulation of the counter (ClockDiv2.SCH).

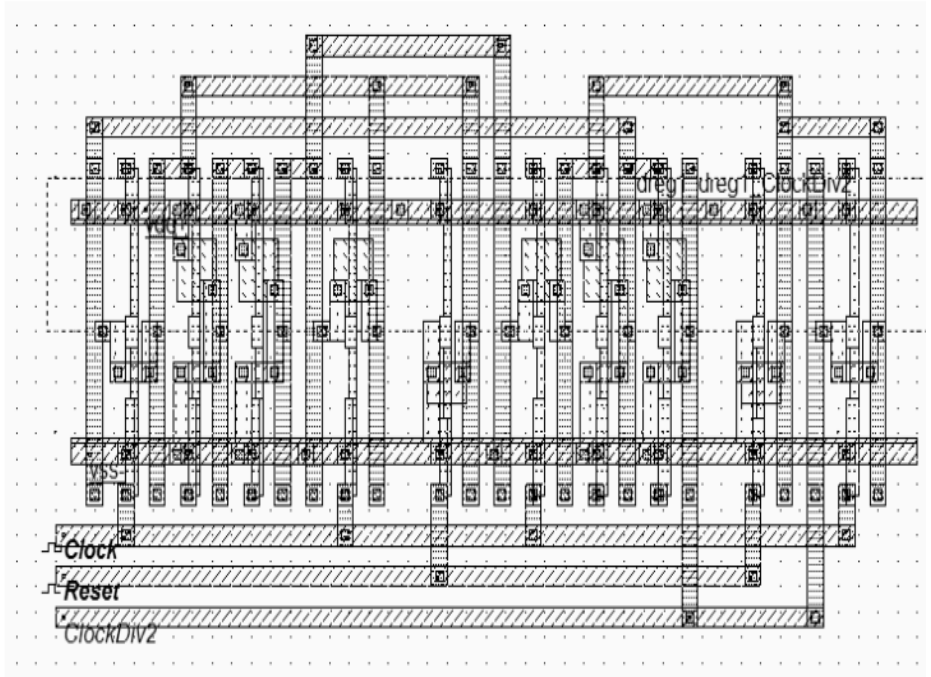


Fig 7-15. Layout of the divider-by-two (ClockDiv2.MSK)

Group Exercise

Task

1. Choose only one digital circuit; identify the digital gates from the verilog code (if you use it to compile for layout)
2. Show the signal path of the circuit
3. Explain the delay based on the timing diagram.
4. Determine the current or power consume by the circuit
5. Repeat (3) and (4) for 180nm (cmos018) and 50 nm (cmos50n) processes.

Knowledge check

WRAP-UP

1. A propagation delay exists due to parasitic capacitance.
2. The main reason to reduce the number of gates in a digital circuit is to reduce power consumption.
3. Spice-based simulation can validate the functional of the digital circuit.

Module 3: submit report
https://drive.google.com/drive/folders/1pJiXCG217wpYqlUotU7CZ2s9T534I9iQ?usp=drive_link